

STAT415 Handouts

Kevin Ross

May 7, 2026

Table of contents

Preface	3
1 Introductory Activity	4
2 Introductory Example	10
2.1 Notes	16
2.1.1 Prior plausibility	16
2.1.2 Simulation	18
2.1.3 Posterior plausibility	20
3 Ideas of Bayesian Reasoning	24
3.1 Notes	28
3.1.1 Prior distribution	28
3.1.2 Likelihood	29
3.1.3 Posterior distribution	31
3.1.4 Posterior is proportional to the product of prior and likelihood	33
3.1.5 Grid approximation	34
4 Interpretations of Probability and Statistics	38
5 Bayes Rule	46
5.1 Notes	52
5.1.1 Two-way table	52
5.1.2 Bayes table, given postgraduate degree	53
5.1.3 Bayes table, given college	54
6 Introduction to Estimation	55
6.1 Notes	63
6.1.1 Likelihood	63
6.1.2 Prior = equally likely, discrete; data = 8/12	63
6.1.3 Prior = triangular, discrete; data = 8/12	66
6.1.4 Prior = “left”, discrete; data = 8/12	68
6.1.5 Prior = equally likely, grid approximation; data = 8/12	71
6.1.6 Prior = triangular, grid approximation; data = 8/12	73
6.1.7 Prior = “left”, grid approximation; data = 8/12	75
6.1.8 Prior = equally likely, grid approximation; data = 80/124	78

6.1.9	Prior = triangular, grid approximation; data = 80/124	80
6.1.10	Prior = “left”, grid approximation; data = 80/124	82
6.1.11	Simulating directly from the posterior distribution	85
7	Introduction to Inference	86
7.1	Notes	91
7.1.1	Normal(0.4, 0.1) prior	91
7.1.2	Posterior based on sample of size $n = 150$ with $\hat{p} = 0.75$	92
7.1.3	Posterior based on sample of size $n = 1502$ with $\hat{p} = 0.75$	95
7.1.4	Conclusions	98
7.1.5	Posterior based on Normal(0.4, 0.05) prior	98
7.1.6	Posterior based on Uniform(0, 0.7) prior	101
8	Introduction to Inference for a Population Mean	103
8.1	Notes	108
8.1.1	Likelihood of single value	108
8.1.2	Posterior based on first observation and discrete prior	109
8.1.3	Posterior based on second observation and discrete prior	110
8.1.4	Posterior based on two observations and discrete prior	111
8.1.5	Posterior based on sample mean of two observations and discrete prior	112
8.1.6	Posterior based on sample of size $n = 2$ and uniform prior	113
8.1.7	Posterior based on sample of size $n = 2$ and triangular prior	115
8.1.8	Posterior based on sample of size $n = 2$ and Normal prior	117
8.1.9	Posterior based on sample of size $n = 208$ and uniform prior	119
8.1.10	Posterior based on sample of size $n = 208$ and triangular prior	120
8.1.11	Posterior based on sample of size $n = 208$ and Normal prior	122
8.1.12	Conclusions	123
9	Comparing Bayesian and Frequentist Analysis	125
10	Introduction to Prediction	131
10.1	Notes	138
10.1.1	Prior predictive probability of success for single trial	138
10.1.2	Prior predictive distribution of sample count of success	139
10.1.3	Posterior distribution - simulation	143
10.1.4	Posterior distribution - calculation	144
10.1.5	Posterior predictive probability of success for single trial	146
10.1.6	Posterior predictive distribution of sample count of success	147
10.1.7	Continuous theta: Prior distribution	151
10.1.8	Continuous theta: Prior predictive distribution	152
10.1.9	Continuous theta: Posterior distribution	153
10.1.10	Continuous theta: Posterior predictive distribution	154
10.1.11	Continuous theta: Posterior distribution after observing large sample	156

10.1.12	Continuous theta: Posterior predictive distribution (after observing large sample)	157
10.1.13	Sleep hours: Prior distribution	158
10.1.14	Sleep hours: Prior predictive distribution	159
10.1.15	Sleep hours: Posterior distribution	160
10.1.16	Sleep hours: Posterior predictive distribution	162
10.1.17	Sleep hours: Posterior predictive distribution after observing large sample	162
11	Predictive Tuning and Checking	165
11.1	Prior predictive tuning	165
11.2	Posterior predictive checking	166
11.3	Notes	168
11.3.1	Prior predictive distribution of sleep hours	168
11.3.2	Posterior predictive checking: Data is singular	169
11.3.3	Posterior predictive checking: Free throws	172
12	Beta-Binomial Model	175
12.1	Notes	181
12.1.1	Prior Beta distributions	181
12.1.2	Posterior Beta distributions	181
12.1.3	Posterior distribution	184
12.1.4	Prior predictive distribution	188
12.1.5	Posterior predictive distribution	190
13	Considering Prior Distributions	193
13.1	Notes	198
13.1.1	Improper Beta(0, 0) prior	198
13.1.2	Prior distribution of odds	198
13.1.3	Prior predictive distribution	199
14	Introduction to Posterior Simulation	201
14.1	Introduction to <code>brms</code>	203
14.1.1	Example: inference for a population proportion	204
14.1.2	Example: inference for a population mean	214
14.1.3	Example: Posterior predictive distribution and checking	222
15	Bayesian Analysis of Poisson Count Data	226
15.1	Notes	232
15.1.1	Discrete prior - single game	232
15.1.2	Discrete prior - multiple games	234
15.1.3	Sample data	235
15.1.4	Grid approximation	237
15.1.5	Mathematical derivation of posterior	238

15.1.6	Posterior simulation with <code>brms</code>	242
15.1.7	Posterior prediction	244
15.1.8	Posterior predictive checking: samples	246
15.1.9	Posterior predictive checking: count 0s	248
16	Normal-Normal Model	250
16.1	Notes	252
17	Introduction to Multi-Parameter models	256
17.1	Notes	260
17.1.1	Discrete grid	260
17.1.2	Prior distribution	262
17.1.3	Prior predictive distribution	263
17.1.4	Grid approximation: $n = 2$	265
17.1.5	Sample data	267
17.1.6	Grid approximation of posterior given sample data	269
17.1.7	Posterior simulation with <code>brms</code>	271
17.1.8	Posterior prediction	278
17.1.9	Posterior predictive checking	280
18	Bayesian Analysis of a Numerical Variable	282
18.1	Notes	287
18.1.1	Prior predictive distribution (Normal likelihood)	287
18.1.2	Sample data	288
18.1.3	Posterior distribution (Normal likelihood)	289
18.1.4	Posterior predictive distribution (Normal likelihood)	293
18.1.5	Posterior predictive checking (Normal likelihood)	295
18.1.6	Posterior distribution (t likelihood)	297
18.1.7	Posterior predictive distribution (t likelihood)	300
18.1.8	Posterior predictive checking (t likelihood)	303
18.1.9	Default <code>brms</code> priors (t likelihood)	304
19	More Bayesian Analysis of a Numerical Variable	308
19.1	Notes	311
19.1.1	Prior predictive distribution (Normal likelihood)	311
19.1.2	Sample data	312
19.1.3	Posterior distribution (Normal likelihood)	314
19.1.4	Posterior predictive distribution (Normal likelihood)	318
19.1.5	Posterior predictive checking (Normal likelihood)	320
19.1.6	Posterior distribution (Skew-Normal likelihood)	322
19.1.7	Posterior predictive distribution (Skew-Normal likelihood)	325
19.1.8	Posterior predictive checking (Skew-Normal likelihood)	327
19.1.9	Default <code>brms</code> priors (Skew-Normal likelihood)	329

19.1.10 Code for Shiny app	333
20 Comparing Two Means	335
20.1 Notes	339
20.1.1 Prior distribution of $\mu_1 - \mu_2$	339
20.1.2 Sample data	340
20.1.3 Posterior distribution	342
20.1.4 Posterior distribution of difference in means	346
20.1.5 Posterior distribution of ratio of means	348
20.1.6 Posterior distribution of effect size	349
21 Introduction to Markov Chain Monte Carlo (MCMC) Simulation	352
21.1 Notes	362
21.1.1 Island simulation: 5 islands, 30 days	362
21.1.2 Island simulation: 5 islands, Many days	364
21.1.3 Island simulation: Many islands, Many days	367
21.1.4 Beta-Binomial	370
21.1.5 Multi-parameter	374
22 Some Diagnostics for MCMC Simulation	381
22.1 Notes	387
23 Bayesian Simple Linear Regression Model	393
23.1 Notes	399
23.1.1 Prior predictive tuning	399
23.1.2 Prior predictive distribution	403
23.1.3 Sample data	405
23.1.4 Frequentist results	407
23.1.5 Posterior distribution	410
23.1.6 Posterior distribution, model using centered BMI	414
23.1.7 Posterior prediction	423
23.1.8 Posterior predictive checking	424
23.1.9 Sensitivity: using <code>brms</code> default prior	427
24 Naive Bayes Classification	434
24.1 Notes	438
24.1.1 Given below average weight	438
24.1.2 Given not below average weight	439
24.1.3 Given 50mm bill	440
24.1.4 Given 50mm bill and below average weight	441
24.1.5 Given 50mm bill and 195mm flipper	442
24.1.6 Data	443
24.1.7 Naive Bayes Classification with <code>e1071</code> package	449

25 Introduction to Bayesian Model Comparison	455
25.1 Odds and Bayes factors	455
25.2 Posterior probability of models	457
25.3 Model averaging	458
25.4 Accounting for model complexity	459
25.5 Sensitivity of model comparison to prior	461
25.6 Hypothesis testing (just don't do it)	463
25.7 Notes	466
25.7.1 Odds and Bayes factors	466
25.7.2 Posterior probability of models	466
25.7.3 Model averaging	467
25.7.4 Accounting for model complexity	469
25.7.5 Sensitivity of model comparison to prior	470
25.7.6 Hypothesis testing (just don't do it)	472
References	475

Preface

Handouts containing examples, exercises, and notes covering selected topics from Cal Poly STAT 415.

Some of the examples use the R package `brms` C. (2017).

Here are some other commonly used packages and settings.

```
library(tidyverse)
library(janitor)
library(kableExtra)
library(viridis)
library(scales)
library(tidybayes)
library(bayesplot)
library(posterior)
library(ggdists)
library(bayesrules)
library(e1071)

bayes_col = c("#56B4E9", "#E69F00", "#009E73", "#CC79A7", "#CC79A7")
names(bayes_col) = c(
  "prior",
  "likelihood",
  "posterior",
  "prior_predict",
  "posterior_predict"
)

bayes_lty = c("dashed", "dotted", "solid")
names(bayes_lty) = c("prior", "likelihood", "posterior")

set.seed(21234)
```